

D027: restrained textual and translation apparatus

Authority physical pages 1–60; article printed pages 103–161

The controlling authority is the 60-page JSTOR scan identified in the publication manifest. Physical page 1 is retained only as provenance and is excluded from both scholarly bodies. The English edition is the source-language edition; the French edition is a faithful translation. Inherited readers, TeX, OCR, and comparator files remain evidentiary until separately verified.

Nine TeX parser-boundary repairs are applied independently in each scholarly edition. Two instances of `\toE` become `\to E` on physical page 9, and one instance of `\inE` becomes `\in E` on physical page 10. Six displayed line breaks immediately followed by a literal opening bracket become `\{ }` on physical pages 35, 55, 56, and 57, preventing TeX from treating the bracketed expression as an optional spacing argument. The genuine `\[2mm]` spacing command on physical page 53 is preserved. These changes restore the intended token boundaries and do not alter any mathematical reading.

Physical page 1 / provenance-only cover

Disposition: provenance only. The cover was inspected directly. The title, authorship, journal series, volume, issue, date, page span, and publisher were retained in the control record. The stable URL, access time, terms-of-use language, copyright/digitization copy, and cover layout were excluded from both scholarly bodies. No authority article text occurs on this page.

Physical page 2 / printed page 103

The publication line, exact title capitalization, and source byline “By P. DELIGNE and G. LUSZTIG” are retained. The inherited TeX/compiled lineage adds a non-source abstract and omits the authority publication line. Its compiled form also rennumbers source citations, prints epsilon in place of the authority’s $\pm$ before R_T^θ , and changes $q \geq 30$ to $q > 30$; all such readings were rejected. The page contains no unrepresentable visual object, so no crop asset was promoted.

Physical page 3 / printed page 104

The running header and folio are anchors only and are excluded. The Introduction’s closing paragraphs, the second-author I.H.E.S. acknowledgement, the complete Table of Contents, and the opening of the conventions are included. In 0.2 the authority prints $H_c^i(X)$ and $H^i(X)$; the inherited candidate’s starred forms were rejected. The tabular leaders were encoded textually without loss requiring an image asset.

Physical page 4 / printed page 105

The page begins mid-sentence, preserving the physical 3/4 transition (“We will” / “make an accidental use”). Authority notation φ_{ji} and σ_i is retained; inherited φ_{ij} and ψ_i normalizations were rejected. The bar automorphism is recorded by the exact printed action $\zeta \mapsto \zeta^{-1}$, rather than by the candidate’s paraphrase. Running header and folio are excluded as anchors.

Physical page 5 / printed page 106

Authority notation $\sigma_{F(i)}^{-1} F \sigma_i$ is retained. The full composite bijection, including arrow directions and the labels σ_i and (e, g) , is preserved in TeX-form text. The printed P , $\bar{L}$, and $X_{\bar{L}}$ readings are retained against candidate normalizations. All diagrammatic structure is representable in text/TeX, so no authority crop was required.

Physical page 6 / printed page 107

The page continues (b3) with fibre P_1 , then opens 1.3 and Definitions 1.4-1.5. The printed class-set notation $W_F^\sim$ and the Borel subgroup σB are retained; inherited W^F and αB substitutions were rejected. The physical page ends with the lead-in to the relative-position diagram; the diagram itself begins on physical 7 and is not imported into this page.

Physical page 7 / printed page 108

The page opens with the relative-position diagram completing Case 1 and retains the authority maps σ , τ , and $\sigma^{(q)}$. In Case 2, the partition $X(w') = X_1 \cup X_2$, the bound $i \geq 0$ in (1.6.1), and the projection $\varphi : X'_2 \rightarrow X_2$ are preserved. Inherited alpha/beta substitutions, the stricter $i > 0$, and the replacement of φ by p were rejected. Both diagrams are fully representable as labelled arrow arrays; no crop was promoted.

Physical page 8 / printed page 109

The cartesian square (1.6.2), exact sequence (1.6.3), trace identities, and Lemma 1.6.4 are retained at their printed page boundary. The first term of (1.6.3) is $H_c^{i-1}(X(sw))$, and the connecting maps are ∂ ; inherited H_c^{i-2} and alpha readings were rejected. No asset is needed for the square or exact-sequence typography.

Physical page 9 / printed page 110

The proof of Lemma 1.6.4 retains $1 \leq j \leq k$ and the printed exchange relation. Section 1.7 preserves the authority plain $E = G/U^*$, $\sigma(T^*, B^*)$, the formula $u(\dot{w}t) = (u\dot{w})t$ in (1.7.2), the direction $\text{pr}_1^* E \rightarrow \text{pr}_2^* E$, and $u\dot{w} = (u\dot{w}_1)\dot{w}_2$. Candidate dot-E notation, strict inequalities, the dot-v substitution in (1.7.2), reversed projections, and the stray v were rejected. All torsor formulas are text/TeX-representable.

Physical page 10 / printed page 111

Authority notation uses plain $E(x, \dot{w})$ and an ambient plain E , while retaining $\tilde{X}(\dot{w})$ and the projection π throughout Definition 1.9 and Example 1.10. The equivariance condition is $\theta(xt) = \theta(x)\theta(t)$, and the finite pushforward decomposes as $\pi_* \tilde{Q}_l = \bigoplus_{\theta} \mathcal{F}_{\theta}$. Inherited dot-E/dot-X notation, projection substitutions, and malformed local-system notation were rejected. No visual asset was required.

Physical page 11 / printed page 112

Equations (1.11.2)-(1.11.4), Corollary 1.12, Lemma 1.13, and Corollary 1.14 are preserved. The set S has no extraneous right factor, (1.13.1) reads $h(T, \text{ad } \dot{w}B) = w^{-1}h(T, B)F(w)$, and the page retains J , W_F , and $\tilde{X}$. Divergent candidate normalizations were rejected; formulas remain fully representable without crops.

Physical page 12 / printed page 113

The general homogeneous-space construction preserves the distinction between $\tilde{E}_0$ and E_0 , and between $\tilde{E}$ and E . Four printed anomalies are retained rather than silently emended: the morphism is introduced as $x : \tilde{E}_0 \rightarrow E_0$ but subsequent formulas use π ; the displayed isomorphism has denominator $S(\pi(e))$; Proposition 1.16 and its proof place $\tilde{e}$ and $\tilde{e}_0$ in plain E and E^F ; and the subsequent map is printed $ge_0 \mapsto g^{-1}Fg$. The printed F-conjugacy action is $w \mapsto w_1 w F(w_1)^{-1}$. The inherited lineage normalizes or collapses these distinctions and changes the action; those readings were rejected. No asset was needed.

Physical page 13 / printed page 114

The page begins with the continuation of Proposition 1.16. Definition 1.17 preserves the three displayed models for $X_{T \subset B}$, including the middle quotient by $B \cap FB$, and uses $\pi : \tilde{X}_{T \subset B} \rightarrow X_{T \subset B}$. Sections 1.18-1.20 retain $\tilde{X}$, $T(w)$, T^{*F} , and $R_{T \subset B}^{\theta} = R^{\theta \circ \text{ad } x}(w)$. The inherited lineage drops the middle model, substitutes malformed “ $\tilde{X}_T \subset B$ ” notation, and replaces π by $\dot{w}$; those readings were rejected. No crop asset was required.

Physical page 14 / printed page 115

The authority uses the superscript quotient notation $G \times^Z T$, and Corollary 1.22 prints H^* , not H_c^* . Proposition 1.23 uses the domain $(T \times \tilde{G})^F$ and the map φ , while 1.24 uses $\varphi : X \rightarrow I$ and cosets B/A . Lemma 1.26 preserves both lower vertical arrows and the identity $(\tilde{G}/\tilde{Z})^F = (G/Z)^F$. The inherited TeX normalizes these to fibre-product subscripts, changes the cohomology and map data, and truncates the lower diagram; those divergences were rejected. The full diagram is representable as a labelled array, so no asset was promoted.

Physical page 15 / printed page 116

The page continues the proof of 1.25 and retains the printed plain- X anomaly in the sentence asserting that $X_{\tilde{T}CB}$ is an induced torsor. Corollary 1.27 reads $R_{TCB}^{\theta\eta} = \theta \otimes R_{TCB}^{\eta}$, and the next display uses H^* and the printed induced-space expression. Chapter 2 retains the source typo “n-dimenssional,” plain E , the action $e \mapsto e \cdot \dot{w}$, the four-term graded isomorphism chain, and the conditions $D_i'' + D_i' = D_{i+1}'$, $D_{n-1}'' + D_1' = V$. The inherited lineage regularizes or changes all of these points. No visual asset was required. The following sentence prints $(G \times^{\tilde{Z}} T)^F$, although the preceding target of induction is $(G \times^Z T)^F$; this superscript anomaly is retained exactly from the authority. The inherited lineage normalizes that occurrence to Z , so its reading was rejected.

Physical page 16 / printed page 117

The page preserves the flag model, congruences, wedge identity, and equations (2.2.1)–(2.2.2). In (2.2.2) the determinant entries are $x_i^{q^{i-1}}$, and the scalar action is by $\lambda \in \mathbb{F}_{q^n}^*$. The Drinfeld form is explicitly displayed as the determinant $\begin{vmatrix} x & x^q \\ y & y^q \end{vmatrix}$. Proposition 2.3 partitions $P(V)$ only through X_{n-1} , and the page ends with the recursive rational-point factor printed by the authority. Candidate readings with q^j , $\mathbb{F}_q^*$, X_n , and a substituted Gaussian-binomial count were rejected. No crop was needed.

Physical page 17 / printed page 118

The proof of Proposition 2.3 retains the recurrence beginning $(q^{nr} - 1)/(q - 1)$, the vanishing range $1 \leq r \leq n - 1$, and the product in $Q = q^r$. Section 2.4 preserves the signed Coxeter permutation and the piecewise point-count formula with q^4 . The inherited lineage supplies a different recurrence, weakens the vanishing range, misstates the permutation, and replaces the piecewise formula by a malformed single expression. Chapter 3 begins at the authority page boundary. All structures are faithfully representable in text/TeX.

Physical page 18 / printed page 119

Theorem 3.2 prints $\text{Tr}(\sigma^*, H_c^*(X))$ on the left and $\text{Tr}(u^*, H_c^*(X^s))$ on the right. The authority also prints the specialization condition $p > 1$, the composite $F^n \circ \sigma$, and $n \geq 1$ in the displayed automorphism identity; these readings are retained without silent emendation. The inherited TeX changes the left trace to u^* , normalizes $p > 1$ to $p > 0$, writes $F \circ \sigma^n$, and changes the displayed range. No crop asset was required.

Physical page 19 / printed page 120

Sections 3.4–3.7 preserve the authority’s coefficient-ring symbol A throughout. The inherited TeX lineage systematically substitutes Λ , while also normalizing the printed l -notation and complex typography; those substitutions were rejected. Equation (3.3), Proposition 3.5, $D^b(A)$, and $R\Gamma_c(Y, \mathcal{F})$ follow the authority page. Every mathematical structure is representable in text/TeX, so no crop asset was promoted.

Physical page 20 / printed page 121

The proof of Proposition 3.7 retains the derived base-change isomorphism, the bullet notation on K_n^* and K_∞^* , and the authority’s notation H_θ^* for the isotypic subspace of compactly supported cohomology. The inherited lineage replaces the isomorphism by an equality, suppresses the bullet notation, and writes $H_{c,\theta}^*$; those readings were rejected. No visual asset was required.

Physical page 21 / printed page 122

Corollary 3.10 and its proof preserve the fixed-locus coproduct with superscript st , the condition $t = t(s, i)^{-1}$, the complete four-line averaging computation, and the final identity without a spurious summation over t . The inherited lineage alters each of these points and also changes H_θ^* ; the frozen authority readings were retained. No crop was needed.

Physical page 22 / printed page 123

The compactification diagram preserves $X \xrightarrow{j} \overline{X}$ above $Y \hookrightarrow \overline{Y}$. The proof of Proposition 3.12 uses $H^*(\overline{X}, j_! \overline{Q}_I)$, the historical remark uses $H^i(X, \mathbb{Z})$, Definition 4.1 prints $R_{T \subset B}^1$, and (4.1.1) names G^{ad} . Contrary inherited normalizations were rejected. The Chapter 4 opening and diagram are faithfully representable without a crop.

Physical page 23 / printed page 124

The page retains α , F^α , the point-count rational function, and the evaluation at $t = \infty$ in (4.1.2). Theorem 4.2 and (4.2.1) preserve their full summation indices, including $g \in G^F$. The printed cross-reference to 3.7 and the expression $\text{ad } g'T$ in the proof of 4.4 are retained exactly; inherited changes to 3.10 and $\text{ad } g'T'$ were rejected. No crop was required.

Physical page 24 / printed page 125

The opening Bruhat inclusion chain is retained in full. Proposition 4.5 prints $w_1 = \overline{\tau}' w \overline{\tau}''^{-1}$ and concludes $w = \overline{\tau}''$; Lemma 4.6 retains the second subscript B' ; Proposition 4.7 retains the superscript s on $X_{T \subset B}^s$. The inherited lineage corrupts or suppresses these features, so its readings were rejected. All structures are representable in text/TeX and no asset was promoted.

Physical page 25 / printed page 126

The Chapter 4 conclusion preserves the superscript s in (4.7.2), the target $X_{\text{ad } gT \subset \text{ad } gB \cap Z^0(s)}$, and the action on $\tilde{X}_{T \subset B}$. At the Chapter 5 opening the authority uses α , not the inherited replacement x , in the pairing $\langle \alpha, h \rangle$, and retains the prime-to- p subscripts in (5.0.1)–(5.2.1). The candidate's malformed Chapter 4 notation and altered duality formula were rejected. All structures are representable in text/TeX; no asset was promoted.

Physical page 26 / printed page 127

The authority gives (5.2.3) with $Y(T) \xrightarrow{F-1} Y(T)$, then prints the full intrinsic five-term sequence with $(F-1)^{-1}$ and the completion-away-from- p notation $Y(T)_{\hat{p}'}(1)$. It defines the character group by the printed superscript $\sim$, retains both dual exact sequences, and writes the norm as $(F^n - 1)/(F - 1) = \sum_0^{n-1} F^i$. The inherited TeX replaces or truncates several of these structures, so its readings were not adopted. No crop was required.

Physical page 27 / printed page 128

Both large commutative diagrams were replayed from the authority with their vertical maps N , ${}^t N$, identities, and the sums $\sum F^i$ and $\sum {}^t F^i$. The authority places $[\theta]$ in the quotient $(X(T) \otimes (Q/\mathbb{Z})_{p'})/\mathbf{W}$, not in a $\mathbf{W}$ -invariant subgroup as the inherited TeX states. The smaller dual-torus square and all statement numbering are preserved. Text/TeX suffices; the comparator-derived crop remains locator-only.

Physical page 28 / printed page 129

Proposition 5.7(ii) retains the count $|Z^{0F}|q^l$, the transpose condition ${}^t(wF)x = x$, and the quotient $\mathbf{T}^*/\mathbf{W}$. Most importantly, the authority says that the trace is zero for $i \neq 0$; the inherited TeX reverses this to $i = 0$, which was rejected. Lemma 5.8, the Poincaré-duality formulas, and the start of 5.10 were checked directly against the pixels. No asset was promoted.

Physical page 29 / printed page 130

Section 5.10 is retained with the authority action on $T \times \tilde{G}$, the printed distinction between $G^F/\pi^F(\tilde{G})$ in (5.10.1) and $G^F/\pi(\tilde{G}^F)$ in Proposition 5.11, and maps defined on T and $T \times T$. The inherited TeX acts on the wrong product, inserts a spurious $\theta(a)^{-1}$, changes the domains, and replaces $F\varphi(t) = \psi(t^{-1}F^n t, \varphi(t))$ by a different conjugation formula; all such changes were rejected. The formulas are faithfully representable without a crop.

Physical page 30 / printed page 131

The norm square is preserved in full. Theorem 5.13 prints reflections s_α and refers to the derived group G' ; the inherited TeX substitutes s_{H_α} and $\tilde{G}$. The authority also retains Z/Z_{red}^0 and the factor $\langle H_\alpha, \theta_1 \rangle \alpha$, where the candidate drops the superscript 0 and substitutes another variable. Remark 5.14 and the page-ending opening of Definition 5.15 are preserved. No image asset was necessary.

Physical page 31 / printed page 132

Page 132 begins in the middle of Definition 5.15(i). The authority reads $V^W = \{0\}$ in Lemma 5.17; the inherited TeX reverses this to $V^W \neq \{0\}$. The scan also preserves the printed spelling “euclidian,” the superscript notation W^a , the cyclic index $i \in \mathbb{Z}/n$, and the conclusion involving $W_0^{a^n}$. No image asset was necessary.

Physical page 32 / printed page 133

The authority retains the prime in $\text{ad } x(Ft) = F' \text{ ad } x(t)$; the inherited TeX drops it in that occurrence. The transpose identities in the proof of Proposition 5.20 and the full duality identity in (5.21.3) were checked directly against the pixels and preserved. No image asset was necessary.

Physical page 33 / printed page 134

The scan prints “an F -stable torus on G ” in (5.21.5), whereas the inherited TeX normalizes this to “in G .” The printed term “unseparable,” the lattice expression for Y_2 , and the complete statements 5.22–5.26 are retained without modernization. No image asset was necessary.

Physical page 34 / printed page 135

At the top of printed page 135 the authority has $Z(\theta)^F$, although the surrounding sentence concerns θ' ; the inherited TeX silently changes it to $Z(\theta')^F$. The frozen English also preserves the printed “containd” in Corollary 5.28 and $\theta(x) = 0$ in the proof of Proposition 5.29. The invariant-space formulation of Theorem 6.2 was retained from the scan. No image asset was necessary.

Physical page 35 / printed page 136

The authority states that $R_T^{\theta^{-1}}$ is dual to R_T^θ ; the inherited TeX instead substitutes $R_T^{\theta'}$. The two geometric diagrams, the compact-support operator $R^i \pi_*$, and the G^F -invariant subspace in the opening of 6.6 were all preserved from the scan. No image asset was necessary.

Physical page 36 / printed page 137

The authority gives the degree-shifted isomorphism $H_c^i(-)(-d) \simeq H_c^{i+2d}(-)$, uses $\bar{S}$, n_g , and the strict filtration condition $w' < w$, and prints the duplication “with with.” The inherited TeX alters or suppresses each of these details and omits part of the invariant/degree structure. All displayed structures were representable faithfully in text and LaTeX, so no image asset was promoted.

Physical page 37 / printed page 138

The authority preserves the product-character notation $\theta\theta'$, the identity-component notation H_w^0 , the printed arrow labels $x^{-1}Fx$ and $t^{-1} \text{ ad } F(w)(t')$, and the final identity $\theta \circ \text{ad } F(w)(\theta') = 1$. The inherited TeX regularizes these as tensor/product-dot notation and expands the maps differently. The Green-function identity (6.9.1) and the opening of Lemma 6.10 were checked directly against the scan. No image asset was required.

Physical page 38 / printed page 139

The authority defines α by the short central correction $\sum_{s \in T^F \cap Z} \theta(s)\theta'(s^{-1})(\dots)$. The inherited TeX replaces this with a different nested centralizer sum and uses complex-conjugation bars where the scan prints inverse character values. The

one-to-one correspondence and each normalization factor in the ensuing calculation were retained from the authority. Two comparator-derived crops remain locator-only; no asset was promoted.

Physical page 39 / printed page 140

The authority reads $\langle R_T^\theta, R_T^1 \rangle = 0$, applies the analogous argument when $|T'^F| \neq 1$, and prints the double-coset equation with label (1.10). At the Chapter 7 opening it uses $X_{T \subset B}$ and states that Steinberg occurs in $R_{T^*}^1 = \text{Ind}_{B^*F}^{G^F}(1)$. The inherited lineage reverses or normalizes each of these readings. No image asset was required.

Physical page 40 / printed page 141

The opening authority computation has no prefactor $1/|G^F|$; after substitution it uses $(-1)^{\sigma(G)-\sigma(T)}/|T^F|$. The scan also prints the typo “the character θ in non-trivial,” includes $|T^F|$ in the denominator of Corollary 7.2, and gives Proposition 7.5 as the full $\text{St}_G(s)^{-1}$ -weighted double sum over T and θ . The inherited TeX changes all four points, most substantially replacing (7.5.1) by a sum of R_T^1 . No image asset was required.

Physical page 41 / printed page 142

The authority’s proof of Proposition 7.5 retains the displayed $|T^F|^{-1}$ factors, the exact torus/character counts, and the full character sums in (7.6.1)–(7.6.3). It does not print the inherited lineage’s extra $|T^F|^{-1}$ in (7.6.2) or $|T^F|$ in (7.6.3). In the second displayed line of the $\langle \mu, \mu' \rangle$ calculation, the scan displays θ without repeating an explicit $\sum_\theta$; this was retained as printed rather than silently supplied. No image asset was required.

Physical page 42 / printed page 143

The authority omits the inherited $|T^F|$ factor from the dimension formula in Proposition 7.9, uses the quotient notation $G^{\text{ad}F}/G_1^F$ in Proposition 7.10, and says in Proposition 7.11 that θ is “a character of T ” before pairing it over T^F . The inherited TeX normalizes these readings to a different quotient and to “a character of T^F .” All formulas were representable faithfully in text and LaTeX, so no image asset was promoted.

Physical page 43 / printed page 144

The authority prints R_T^1 in (7.11.2), gives the short counting identity without the inherited extra $|G^F|/|Z^0(s)^F|$ factor, retains inverse character values $\theta(g^{-1}sg)^{-1}$ and $\theta(t)^{-1}$, and places no sign in (7.12.1) while placing $(-1)^{\sigma(G)-\sigma(T)}$ in (7.12.2). The inherited TeX changes each of these readings and also regularizes the printed quotient expression involving $X_{T \subset B}/G^F$. The authority formulas were fully representable in text and LaTeX; no image asset was promoted.

Physical page 44 / printed page 145

The authority uses ρ'' , not ρ' , in the zero-norm identity following (7.12.5), prints $|W(T)^F|^{-1}$ in Remark 7.13, and distinguishes the unsigned formula (7.14.1) from the signed formula (7.14.2). The inherited lineage substitutes $|T^F|^{-1}$, carries the sign into (7.14.1), and uses the wrong prime in the norm computation. No image asset was required.

Physical page 45 / printed page 146

The authority preserves the two Corollary 7.15 formulas and opens Chapter 8 with the notation $R_{T,P}^\theta$ as the image of $R_{\pi(T)}^\theta$. Proposition 8.2 reads $R_T^\theta = \text{Ind}_{P^F}^{G^F}(R_{T,P}^\theta)$, and the geometric decomposition uses $\tilde{X}_{T \subset B}$ and $\tilde{X}(P')$. The inherited TeX mangles these into product-dot notation and a self-referential induction formula. All displayed structures were representable without a promoted crop.

Physical page 46 / printed page 147

The authority identifies the regular representation as $\text{Ind}_e^{G^F}(1)$, gives (8.3.1) without an inherited $|U_P^F|$ multiplier, and pairs (8.3.2) with $\text{Ind}_{U_P^F}^{G^F}(1)$, not with induction from P^F . It also preserves the Chapter 9 opening and the beginning of

Definition 9.2. These authority readings override the divergent inherited TeX; no asset was needed.

Physical page 47 / printed page 148

The authority uses $\Psi(\dot{w})$, oriented from $\text{pr}_1^* E_\lambda$ to $\text{pr}_2^* E_{w^{-1}(\lambda)}$, and the commutative square carries the same pr_1 -to- pr_2 orientation. The inherited TeX renames the map, reverses the projections in the isomorphism and rational-section bundle, and adds dotted bundle notation absent from the scan. The diagram was preserved in LaTeX; no image asset was promoted.

Physical page 48 / printed page 149

The authority prints μ in the inequality defining $D(C, -wC)$ and then λ in (9.5.1), a local symbol inconsistency that the inherited TeX silently normalizes. It also gives the specific lift $\dot{w} = \dot{w}_{i-1} \dot{s}_i \dot{w}'_i$, the corresponding composition order for Ψ , an upper-right asterisk in the displayed triangular matrix, and the pr_1 -then- pr_2 line-bundle order in 9.6. All were retained from the authority; no asset was required.

Physical page 49 / printed page 150

The authority continues the line-bundle argument with λ and $\Psi(\dot{w})$, prints (9.6.2) as $F^* \mu - w\mu \in C^0$, and then states locally that the proof treats $q \geq h$, although Theorem 9.7 says “larger than” h . The inherited Part 1 lineage instead uses χ and $T(\dot{w})$, changes (9.6.2) to membership in $-C$, and replaces $q \geq h$ by $q > h$. The authority’s $\tilde{X}(\dot{w})$, isomorphism arrows, and the exact opening of 9.10 were retained. No image asset was required.

Physical page 50 / printed page 151

The authority uses Γ for the graph inclusion and preserves the displayed fibre-product square. Lemma 9.11 literally prints $\overline{O}(s, \dots, s)$ and $\overline{X}(s, \dots, s_n)$, rather than the inherited normalization to $s_1, \dots, s_n$. In 9.12 it uses $\tilde{X}(\dot{w})$, a completed $\tilde{Z}(1)$, the map $t \mapsto B_i(t)$, Ψ -notation, and formulas containing $H_{-\alpha}(t)$ and $H_{w_{i-1}(\alpha_i)}(t)$, without the inherited added $\alpha_i(t)$. The diagram and all formulas were representable in LaTeX; no crop was promoted.

Physical page 51 / printed page 152

The authority continues (9.12.1) with $F(u_0) = \Psi_0(u_0)$, prints (9.12.2) with $H_{w_{i-1}(\alpha_i)}(t)$, and uses the prime-to- p completions $\hat{Z}_{p'}(1)$ and $Y_{p'}^\wedge(1)$. Remark 9.15.2 cites (5.17), not the inherited (5.25), and ends the physical page mid-sentence after “the proof of the Induction Theorem”. The inherited Part 1 file instead terminates the remark there, inserts a fourteen-item bibliography and `\end{document}`, and thereby creates a false article EOF. Those materials were rejected; the frozen authority page remains continuous with physical 52. No asset was required.

Physical page 52 / printed page 153

The authority begins with the direct continuation “8.2 show ...”, then states Theorem 9.16 and Lemma 9.17; it contains no title, bibliography, or restart. This page is the decisive repair of the inherited split: Part 2 supplies the continuation only as unverified lineage, while Part 1’s prior bibliography and EOF are not article content. Formula (9.17.1), the completed local rings $\hat{D}_P$, the equations $P_i = x_i(1 + Q_i)$, and the four-line intersection-multiplicity computation were retained from the scan. No image asset was required.

Physical page 53 / printed page 154

The authority completes Lemma 9.17, gives Proposition 9.18 with the factor $|Z^0 Z^0(s)^F| q^{l(Z^0(s))} |Z(s)^F| / |Z^0(s)^F|$, preserves $\theta(s)^{-1}$ in the virtual character and $\theta(s)$ in Corollary 9.19, and states Lemma 9.20 before opening Chapter 10. The exact map $(G^F)^\sim \rightarrow \mathfrak{S}$ and the chapter boundary were preserved. The Part 2 witness is comparison-only and does not supersede the authority. No image asset was required.

Physical page 54 / printed page 155

The authority uses the bullet-subscript notation U_*^* , not the inherited star-subscript surrogate, throughout 10.2 and 10.5. It defines $\Gamma_G = \text{Ind}_{U_*^F}^{G^F}(\chi)$, the regular-unipotent class function Δ_G , Proposition 10.3 and formula (10.3.1), Remark 10.4, and the opening of the proof in 10.5. The authority also literally writes that χ induces a character of U_i^* in the definition of $\mathcal{C}(J)$. All notation and displays were retained without a promoted crop.

Physical page 55 / printed page 156

The authority continues 10.5 with formulas (10.5.1)–(10.5.2), the regular-unipotent count, and Lemma 10.6. It retains the bullet-subscript U_*^F , $\theta \in T^{\vee F}$, and the full $M_x + M'_x$ orthogonal decomposition. The inherited Part 2 lineage normalizes the bullet subscript to U_*^{*F} ; that normalization was rejected. No image asset was required.

Physical page 56 / printed page 157

The authority completes Lemma 10.6 and opens Theorem 10.7. The reciprocal-norm identity is a global sum over all $(T, \theta) \bmod G^F$, with no inherited $[\theta] = x$ restriction. The determinant identity, the definition $F^* = q\tau^{-1}$, formulas (10.7.1)–(10.7.4), and the continuous Chapter 10 numbering were preserved. No image asset was required.

Physical page 57 / printed page 158

The authority gives (10.7.5)–(10.7.9) and the dimension argument. It literally uses G^F -conjugacy classes in the general case, then G^{*F} -conjugacy classes on the dual side; the first displayed dimension surrogate is induced to G^{*F} , while the second is induced to G^F . These authority-controlled targets supersede the inherited TeX normalizations. No image asset was required.

Physical page 58 / printed page 159

The authority completes Theorem 10.7, proves the required inner products for Γ_G and Δ_G , and states Corollary 10.8, Remark 10.9, and the opening of Proposition 10.10. It prints a second formula label (10.7.8) for the parabolic-induction vanishing; the inherited lineage silently rennumbers it (10.7.10). The printed duplicate was retained as an authority feature. No image asset was required.

Physical page 59 / printed page 160

The authority completes Proposition 10.10 and opens Chapter 11, “Suzuki and Ree groups,” on printed page 160. It preserves the paired signs involving $\sigma(L)$, $\sigma(G)$, $\sigma(T)$, the exceptional isogeny F' , the notation $\tilde{X}_{TCB}$, and the uninterrupted article topology. The inherited Part 2 scaffold remains comparison-only. No image asset was required.

Physical page 60 / printed page 161

The authority completes Chapter 11 and the article. It includes the $|T^{F'}| > 1$ condition, the exceptional ${}^2F_4(\sqrt{2})$ limitation, the Added in Proof dated 2nd Dec. 1975, both institutional addresses, exactly fifteen references, the unclosed parenthesis in reference [8], and “(Received June 6, 1975).” The remainder of the page is blank and physical page 60 is the true EOF. No inherited bibliography or synthetic terminus was promoted.